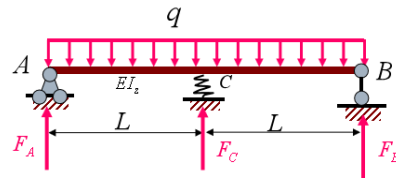


Submission: Please submit the scanned/typed document to Canvas.

1. As shown, a simply supported beam AB with Young's modulus E and moment of inertia I_z carries a uniform distributed load q . An elastic (spring) support with stiffness k is added at the midpoint C . If the bending moment at section C is zero, determine the spring stiffness k .



Solution:

Equilibrium equation

$$F_A + F_C + F_B = 2qL$$

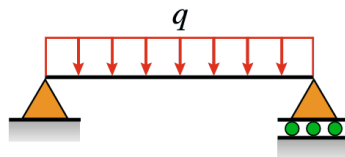
$$-F_A L + F_B L = 0$$

Compatibility equation

$$F_C = k\delta_C$$

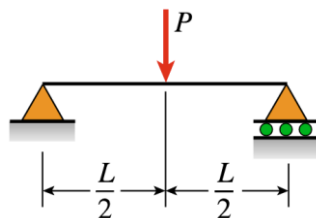
Using method of superposition to calculate the deflection at point C:

(1) Distributed load q



$$\delta_{C1} = \frac{5q(2L)^4}{384EI_z}$$

(2) Concentrated load F_C



$$\delta_{C2} = \frac{-F_C(2L)^3}{48EI_z}$$

The compatibility equation becomes

$$F_C = k \left(\frac{5q(2L)^4}{384EI_z} - \frac{F_C(2L)^3}{48EI_z} \right) \Rightarrow F_C = \frac{5kqL^4}{24EI + 4kL^3}$$

Together with equilibrium equation

$$F_A = qL - \frac{1}{2}F_C = qL - \frac{5kqL^4}{48EI + 8kL^3}$$

The bending moment at point C

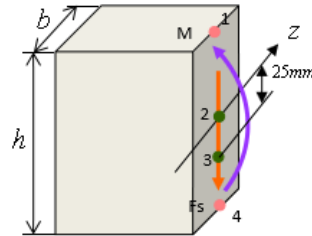
$$M_C = F_A L - \frac{1}{2}qL^2 = \frac{1}{2}qL^2 - \frac{kqL^5}{48EI + 8kL^3}$$

$M_C = 0$ gives

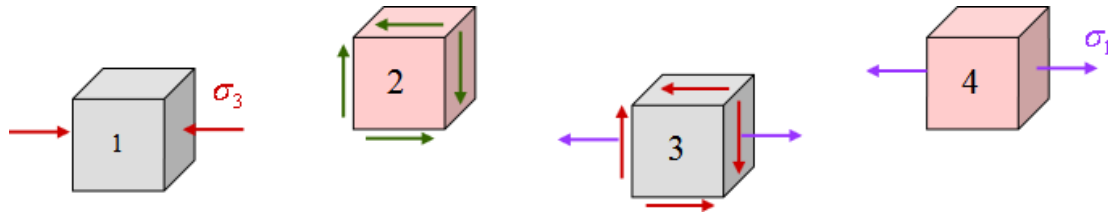
$$k = \frac{24EI}{L^3}$$

2. For a rectangular cross-section beam, the shear force on the section is $F_s = 120$ kN, and the bending moment is $M = 10$ kN · m. Draw the stress element at points 1, 2, 3, and 4 on the cross-section to represent their stress states and determine the principal stresses at each point. $b = 60$ mm, $h = 100$ mm. (Hint: $\tau = \frac{VQ}{Ib} = \frac{V}{2I} \left(\frac{h^2}{4} - y^2 \right)$)

point. $b = 60$ mm, $h = 100$ mm. (Hint: $\tau = \frac{VQ}{Ib} = \frac{V}{2I} \left(\frac{h^2}{4} - y^2 \right)$)



Solution:



$$I_z = \frac{bh^3}{12} = 500 \text{ cm}^4$$

At point 1:

$$\sigma = -\frac{My}{I_z} = -\frac{10 \times 10^3 \times 50 \times 10^{-3}}{500 \times 10^{-8}} \text{ Pa} = -100 \text{ MPa}$$

$$\tau = 0$$

$$\sigma_1 = \sigma_2 = 0, \quad \sigma_3 = -100 \text{ MPa}$$

At point 2:

$$\sigma = 0$$

$$\tau_{\max} = \frac{3 F_s}{2 A} = \frac{3}{2} \frac{120 \times 10^3}{60 \times 100 \times 10^{-6}} \text{ Pa} = 30 \text{ MPa}$$

$$\sigma_1 = 30 \text{ MPa}, \quad \sigma_2 = 0, \quad \sigma_3 = -30 \text{ MPa}$$

At point 3:

$$\sigma = -\frac{My}{I_z} = \frac{10 \times 10^3 \times 25 \times 10^{-3}}{500 \times 10^{-8}} \text{ Pa} = 50 \text{ MPa}$$

$$\tau = \frac{F_s Q}{I_z b} = \frac{120 \times 10^3}{2 \times 500 \times 10^{-8}} \left(\frac{0.1^2}{4} - 0.025^2 \right) = 22.5 \text{ MPa}$$

$$R = \sqrt{\left(\frac{\sigma}{2} \right)^2 + \tau^2} = 33.6 \text{ MPa}$$

$$\sigma_1 = \frac{\sigma}{2} + R \approx 58.6 \text{ MPa}, \quad \sigma_2 = 0, \quad \sigma_3 = \frac{\sigma}{2} - R \approx -8.6 \text{ MPa}$$

At point 4:

$$\sigma = -\frac{My}{I_z} = \frac{10 \times 10^3 \times 50 \times 10^{-3}}{500 \times 10^{-8}} \text{ Pa} = 100 \text{ MPa}$$

$$\tau = 0$$

$$\sigma_1 = 100 \text{ MPa}, \quad \sigma_2 = \sigma_3 = 0$$

3. When water inside a cast-iron pipe freezes in winter, the pipe cracks due to the expansion of ice, but the ice itself is not damaged. This is because (**B**).

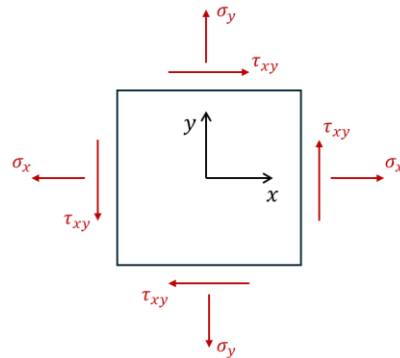
A. The strength of ice is higher than that of cast iron

B. The ice is in a triaxial compressive stress state

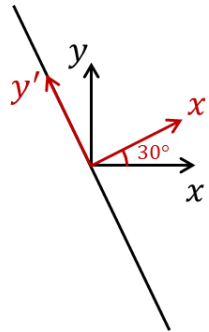
C. The temperature of the ice is higher than that of cast iron

D. The stress in the ice is zero.

4. A structural element is subjected to a uniform plane-stress state denoted by σ_x , σ_y , τ_{xy} in the x - y coordinates. Assume $\sigma_x = -7$ MPa, $\sigma_y = 1$ MPa, and $\tau_{xy} = -3$ MPa.



(1) Please determine the normal and shear stresses acting on the plane whose outward normal, x' -axis, is oriented at an angle to x -axis (see the figure below).

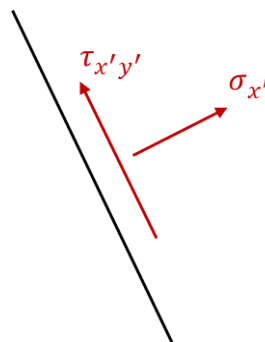


Solution:

$$\theta = 30^\circ$$

$$\text{Normal stress: } \sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta = -5 - \frac{3\sqrt{3}}{2} \text{ MPa.}$$

$$\text{Shear stress: } \tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta = 2\sqrt{3} - \frac{3}{2} \text{ MPa.}$$



(2) Please find the planes that are free of normal stress and determine the shear stress on these planes.

Solution (Method I):

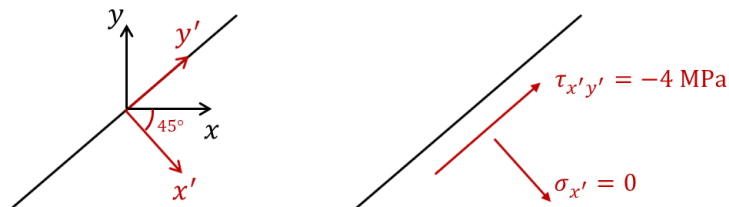
Find θ such that $\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta = -3 - 4 \cos 2\theta - 3 \sin 2\theta =$

0. By noting $\cos^2 2\theta + \sin^2 2\theta = 1$, we find two solutions, θ_1 and θ_2 .

θ_1 satisfies $\sin 2\theta_1 = -1$ and $\cos 2\theta_1 = 0 \rightarrow \theta_1 = -\frac{\pi}{4} = -45^\circ$.

θ_2 satisfies $\sin 2\theta_2 = \frac{7}{25}$ and $\cos 2\theta_2 = -\frac{24}{25} \rightarrow \theta_2 = \frac{1}{2} \left(\pi - \arcsin \frac{7}{25} \right) = 1.4289 = 81.87^\circ$.

When $\theta = \theta_1 = -45^\circ$, $\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta = 0$, $\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta = -4 \text{ MPa}$.

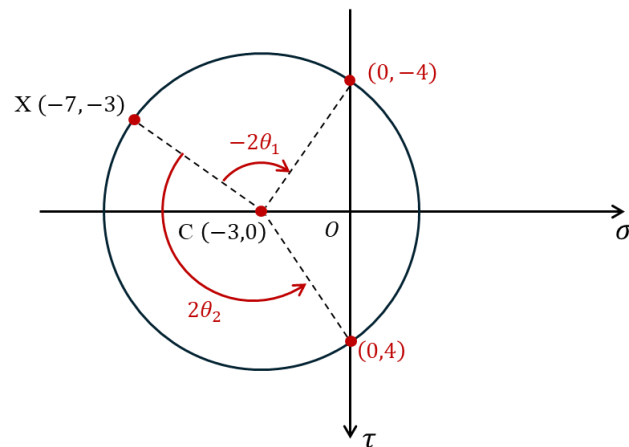


When $\theta = \theta_2 = 81.87^\circ$, $\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta = 0$, $\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta = 4 \text{ MPa}$.

**Solution (Method II):**

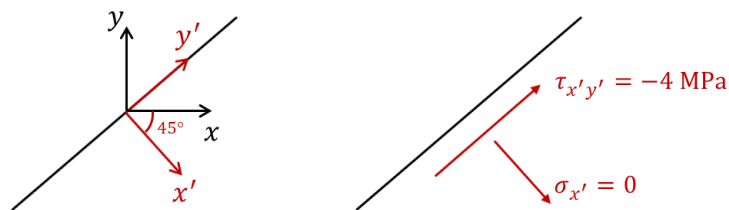
We first construct the Mohr's circle for the specific stress state.

Radius: $R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 5 \text{ MPa}$; Center: $(\sigma_{\text{avg}}, 0)$, $\sigma_{\text{avg}} = -3 \text{ MPa}$.



The τ -intercepts correspond to the planes that are free of normal stress: $(0, -4 \text{ MPa})$ and $(0, 4 \text{ MPa})$. The corresponding rotation angles (positive if counterclockwise) can be calculated from the geometry: $\theta_1 = -45^\circ$ and $\theta_2 = 81.87^\circ$.

When $\theta = \theta_1 = -45^\circ$, $\sigma_{x'} = 0$, $\tau_{x'y'} = -4 \text{ MPa}$.



When $\theta = \theta_2 = 81.87^\circ$, $\sigma_{x'} = 0$, $\tau_{x'y'} = 4 \text{ MPa}$.



- (3) Please determine the principal stresses. Show them on a sketch of a properly oriented element.

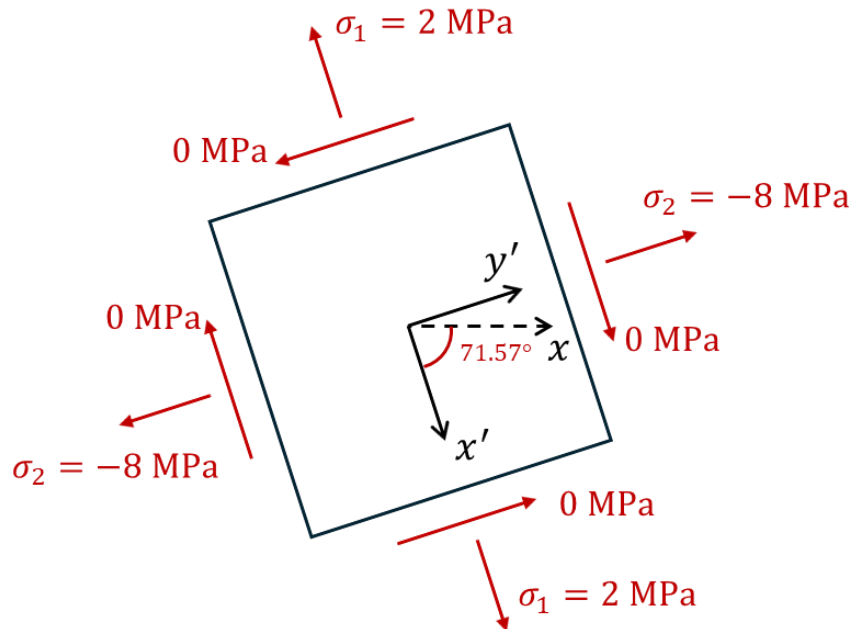
Solution (Method I):

Find θ_p such that $\left. \frac{d\sigma_{x'}}{d\theta} \right|_{\theta_p} = 0$. We have two solutions, θ_{p1} and θ_{p2} .

θ_{p1} satisfies $\sin 2\theta_{p1} = \frac{\tau_{xy}}{R} = -\frac{3}{5}$ and $\cos 2\theta_{p1} = \frac{\sigma_x - \sigma_y}{2R} = -\frac{4}{5} \rightarrow \theta_{p1} =$

$$\frac{1}{2} \left(a \sin \frac{3}{5} - \pi \right) = -1.2490 = -71.57^\circ. \rightarrow \sigma_1 = \sigma_{x'} = 2 \text{ MPa}, \tau_{x'y'} = 0 \text{ MPa}.$$

$$\theta_{p2} = \theta_{p1} + \frac{\pi}{2} = 0.3218 = 18.43^\circ \rightarrow \sigma_2 = \sigma_{x'} = -8 \text{ MPa}, \tau_{x'y'} = 0 \text{ MPa}.$$

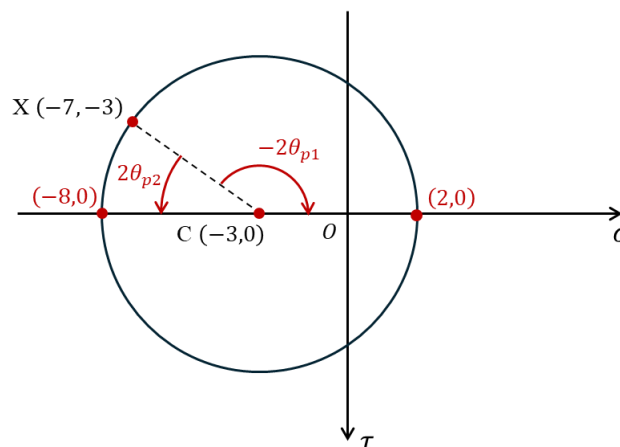


Solution (Method II):

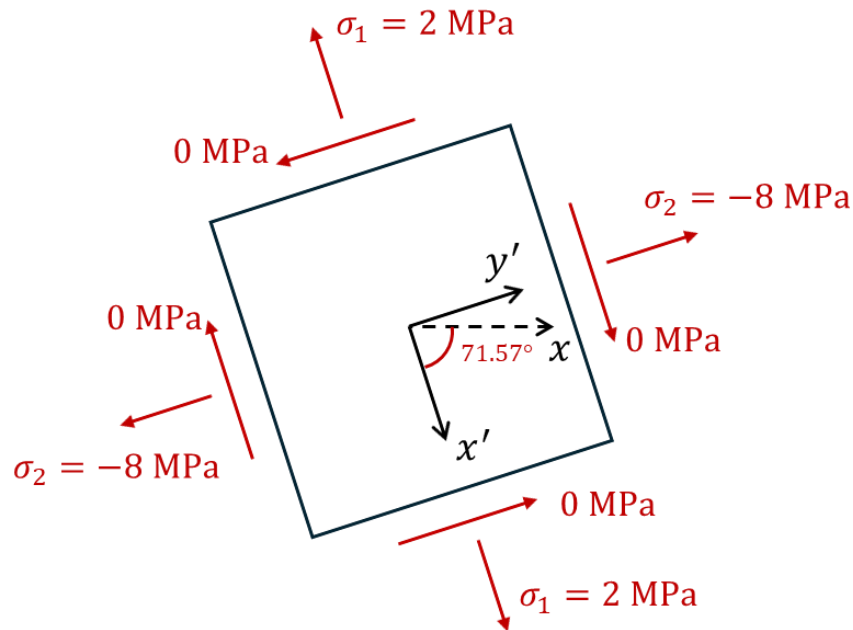
We first construct the Mohr's circle for the specific stress state.

$$\text{Radius: } R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 5 \text{ MPa; Center: } (\sigma_{\text{avg}}, 0), \sigma_{\text{avg}} = -3 \text{ MPa}.$$

The σ -intercepts correspond to the principal planes: (2 MPa, 0) and (-8 MPa, 0). The corresponding rotation angles (positive if counterclockwise) can be calculated from the geometry: $\theta_{p1} = -71.57^\circ$ and $\theta_{p2} = 18.43^\circ$.



The two principal stresses and planes are shown below:



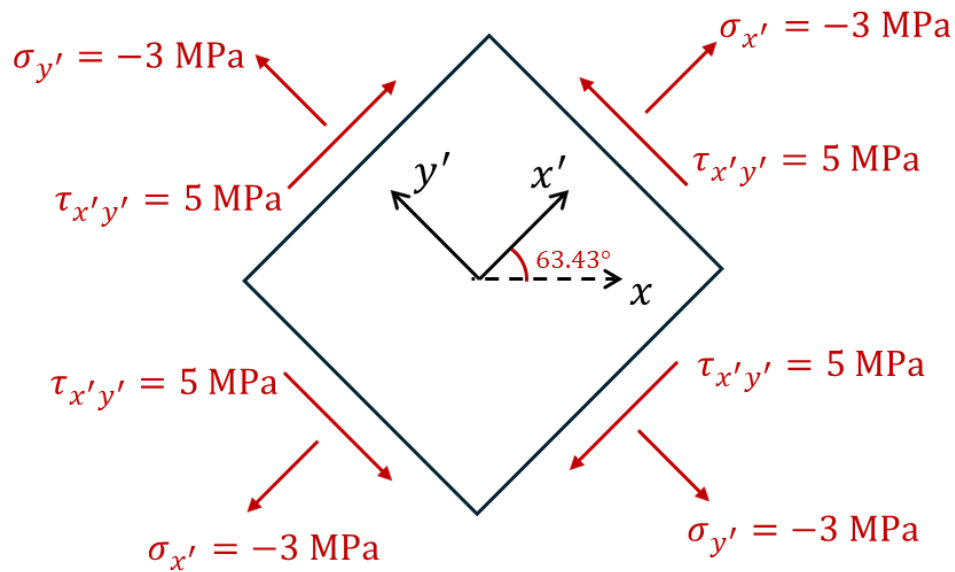
- (4) Please determine the maximum shear stress and the associated normal stress. Show them on a sketch of a properly oriented element.

Solution (Method I):

Find θ_s such that $\left. \frac{d\tau_{x'y'}}{d\theta} \right|_{\theta_s} = 0$. We have two solutions, θ_{s1} and θ_{s2} .

θ_{s1} satisfies $\sin 2\theta_{s1} = -\frac{\sigma_x - \sigma_y}{2R} = \frac{4}{5}$ and $\cos 2\theta_{s1} = \frac{\tau_{xy}}{R} = -\frac{3}{5}$. $\rightarrow \theta_{s1} = \frac{1}{2} \left(\pi - \arcsin \frac{4}{5} \right) = 1.1071 = 63.43^\circ$. $\rightarrow \tau_{s1} = \tau_{x'y'} = R = 5 \text{ MPa}$, $\sigma_{x'} = \sigma_{\text{avg}} = -3 \text{ MPa}$.

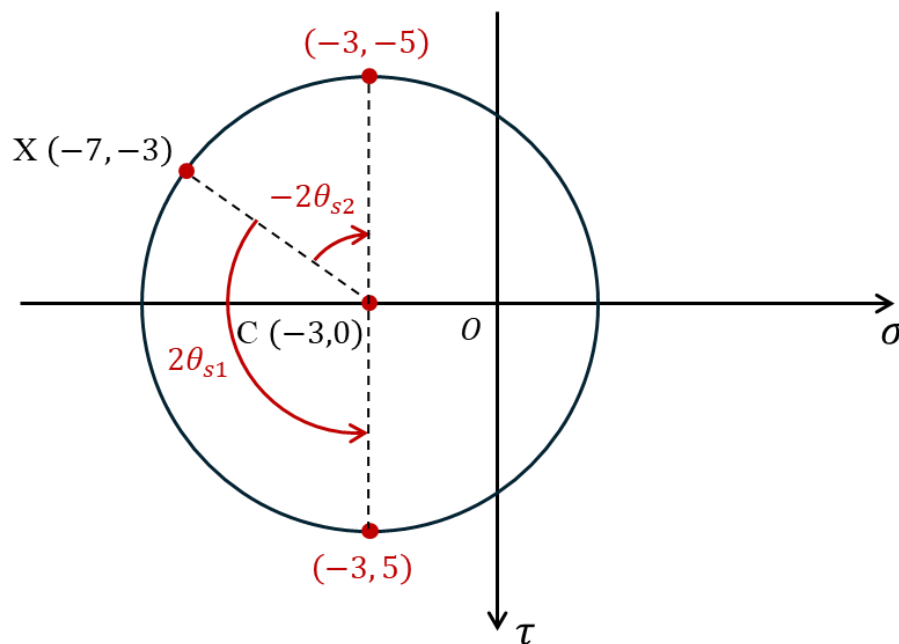
$\theta_{p2} = \theta_{p1} - \frac{\pi}{2} = -0.4637 = -26.57^\circ \rightarrow \tau_{s2} = \tau_{x'y'} = -R = -5 \text{ MPa}$, $\sigma_{x'} = \sigma_{\text{avg}} = -3 \text{ MPa}$.

**Solution (Method II):**

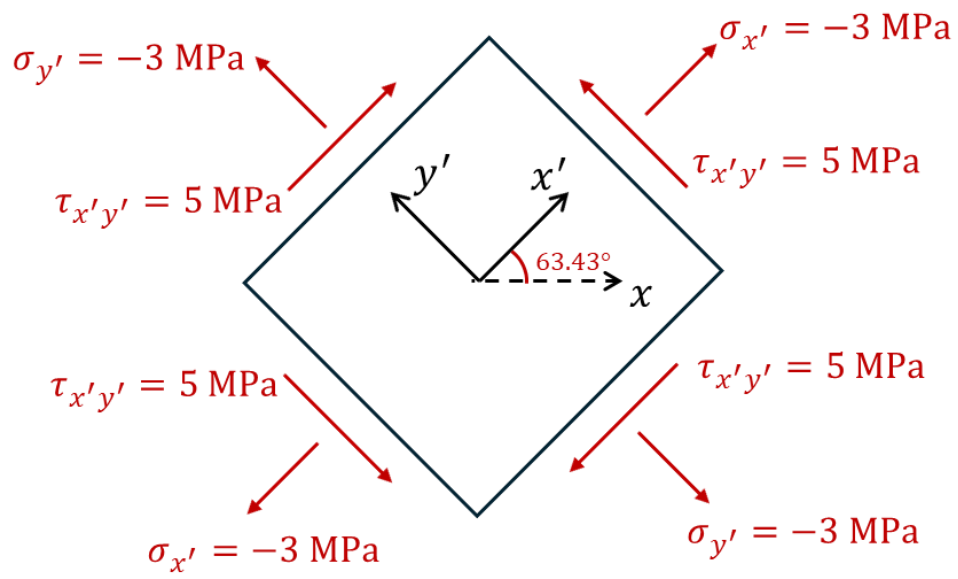
We first construct the Mohr's circle for the specific stress state.

Radius: $R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 5 \text{ MPa}$; Center: $(\sigma_{\text{avg}}, 0)$, $\sigma_{\text{avg}} = -3 \text{ MPa}$.

The points corresponding to the maximum shear stress are $(-3 \text{ MPa}, 5 \text{ MPa})$ and $(-3 \text{ MPa}, -5 \text{ MPa})$. The rotation angles (positive if counterclockwise) can be calculated from the geometry: $\theta_{s1} = 63.43^\circ$ and $\theta_{s2} = -26.57^\circ$.



The stresses on the two planes are shown below:



5. During a test of an airplane wing, the strain gage readings from a 45° rosette (see figure) are as follows:

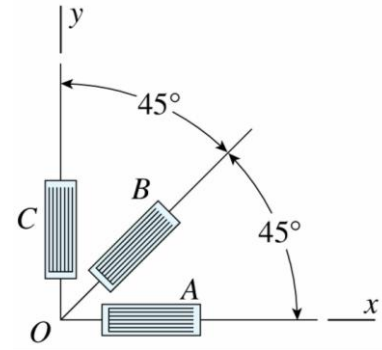
$$\text{gage A, } 520 \times 10^{-6}$$

$$\text{gage B, } 360 \times 10^{-6}$$

$$\text{gage C, } -80 \times 10^{-6}$$

Determine the principal strains and maximum shear strains and show them on sketches of properly oriented elements.

(Hint: For a 45° rosette, $\varepsilon_x = \varepsilon_A$, $\varepsilon_y = \varepsilon_C$, $\gamma_{xy} = 2\varepsilon_B - \varepsilon_A - \varepsilon_C$)



Solution:

$$\varepsilon_x = \varepsilon_A = 520 \times 10^{-6}$$

$$\varepsilon_y = \varepsilon_C = -80 \times 10^{-6}$$

$$\gamma_{xy} = 2\varepsilon_B - \varepsilon_A - \varepsilon_C = 280 \times 10^{-6}$$

The principal strains

$$\varepsilon_1 = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} = 551 \times 10^{-6}$$

$$\varepsilon_2 = \frac{\varepsilon_x + \varepsilon_y}{2} - \frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} = -111 \times 10^{-6}$$

Rotation angle

$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \frac{280}{600}$$

$$\theta_{p1} = 12.5^\circ, \quad \theta_{p2} = 102.5^\circ$$

The maximum shear strain

$$\frac{\gamma_{\max}}{2} = \frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} \Rightarrow \gamma_{\max} = 662 \times 10^{-6}$$

$$\theta_{s1} = \theta_{p1} + 45^\circ = 57.5^\circ$$

The minimum shear strain

$$\frac{\gamma_{\min}}{2} = -\frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} \Rightarrow \gamma_{\min} = -662 \times 10^{-6}$$

$$\theta_{s2} = \theta_{p2} + 45^\circ = 147.5^\circ$$